

Observations April 16, 2026

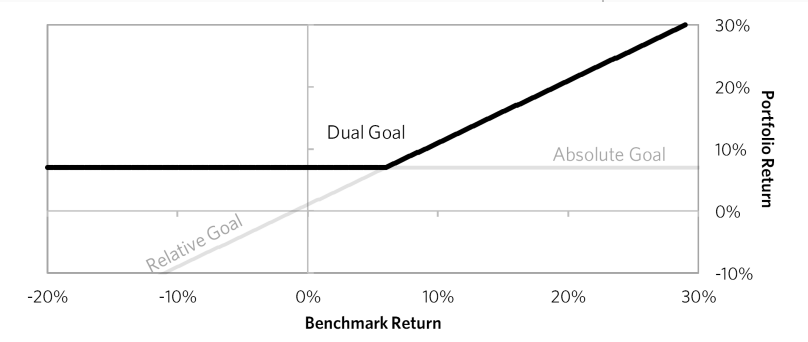
The Portfolio Engineering Trilemma

Many investors face the challenge of balancing absolute returns, relative returns, and investment constraints. We share how we think about navigating this trilemma by building the best possible total portfolio.

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As more institutional investors move toward a Total Portfolio Approach (TPA), we have increasingly heard from our clients that the tension between absolute and relative return goals is becoming more acute. Most investors need to hit some absolute return threshold—whether to fund liabilities, preserve purchasing power, or maintain future deployment capacity. At the same time, sustained underperformance versus passive alternatives creates governance headaches and credibility problems with stakeholders.

We find it helpful to represent those dual goals as a single objective function, one that we suspect would look familiar to many institutional investors. Below, we show an example of this objective function for an investor seeking to generate the better of either a 7% total return or 100bps of outperformance relative to a benchmark.



When we stare at the challenge of meeting these dual goals, we see a trilemma in portfolio engineering: it's exceedingly difficult to consistently produce target returns, beat a benchmark, and operate within tight constraints. One intuitive approach to constructing a portfolio—building something that looks like the benchmark but slightly better—comes with the risk of not meeting absolute return goals when the benchmark inevitably suffers.

In this Observations, we share how we're wrestling with this trilemma and what our approach would be for building the best portfolio that can achieve desired absolute and relative returns. As we see it:

- Give Feedback
- Saved to My Shelf
- Share

The Trilemma

Taking Stock of the Goal(s)

Shaping Returns Toward the Goal(s)

Looking Different by Design

MORE ON Portfolio Construction

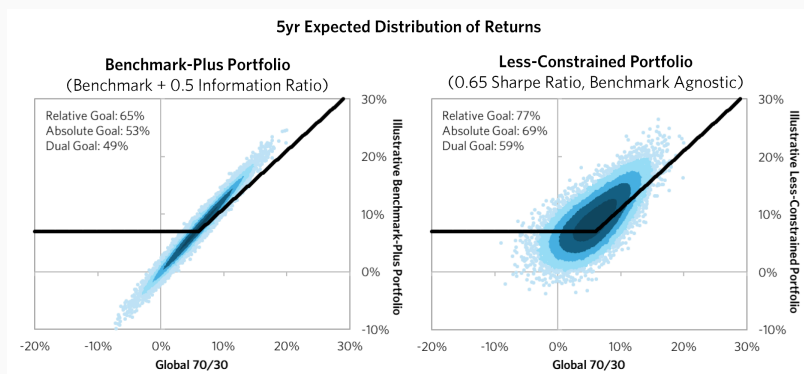
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Karen Karniol-Tambour on Investing in a More Edgingly World

Karen Karniol-Tambour discusses the investment implications of a world increasingly organized around resilience, industrial...

- **Benchmark-plus approaches that optimize more for relative returns than absolute returns will frequently disappoint on absolute return goals.** Even with strong value-add, portfolios built to closely track benchmarks inherit the benchmark's fundamental vulnerabilities. For a portfolio targeting an illustrative 7% absolute return, we estimate that a strong benchmark-plus portfolio (0.5 information ratio with 3% tracking error) would miss the absolute return target nearly half the time, including over longer periods like a decade.
- **A less-constrained approach to portfolio construction can improve the odds of achieving both goals simultaneously.** Portfolios that target a risk level and optimize across all available return sources—rather than anchoring to a benchmark—can significantly improve chances of success, particularly in the long term. For example, constructing a higher-ratio portfolio (0.65 Sharpe ratio) at similar risk leads to a roughly two-thirds chance of meeting ambitious dual investment goals compared to the roughly coin-flip odds of a benchmark-plus approach.
- **Materially improving outcomes can mean tolerating extended periods of divergence from benchmarks and peer portfolios.** This is most difficult during periods when the market-cap portfolio is performing well—precisely when governance pressures to look like everyone else are strongest. But we think building a better total portfolio is ultimately the best way to serve the goals most organizations hold.

Against a dual objective, we can visualize these two broad paths down which investors might guide their portfolios—a benchmark-plus approach or a less-constrained total portfolio. Particularly over longer time frames, we think a less-constrained TPA holds the promise of more consistently achieving the multiple goals that most investors must balance.

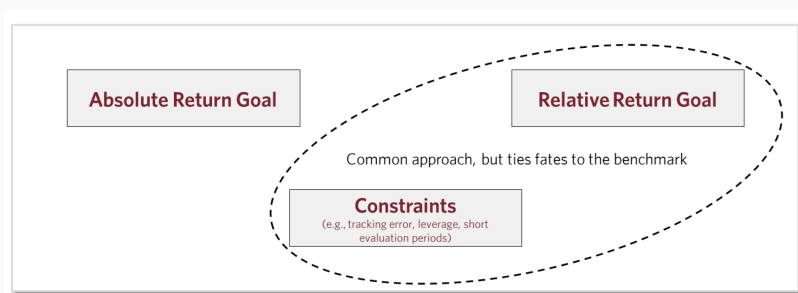


There can be no guarantee that any expected performance can or will be achieved, and expected performance should not be solely relied upon in making any investment decision. Note: "Global 70/30" refers to a portfolio of 70% global equities and 30% global bonds.

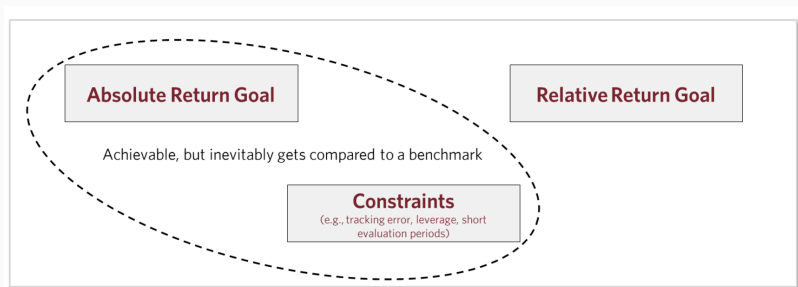
We recognize that our less-constrained approach doesn't fit every investor's circumstances. However, even where a material change to a portfolio is not possible, we hope that understanding these trade-offs explicitly can help clarify the implicit cost of constraints and provide a framework for making informed decisions about portfolio construction.

The Trilemma

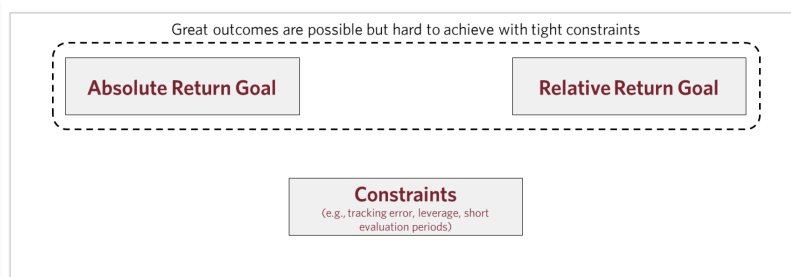
Any portfolio engineering exercise starts with goals and constraints. Across a wide range of investor circumstances, competing goals—both explicit and implicit—quickly come into tension, as do the limits imposed by constraints. The trilemma arises at that intersection. **When we think about building a portfolio to achieve these goals, we could stay within tight constraints and build an allocation designed to reliably beat a benchmark, but this portfolio would likely experience many periods of unacceptably low absolute returns.**



On the other hand, **we could design a portfolio that is better aligned with an absolute goal while remaining within constraints—but those portfolios seemingly always wind up being compared to a benchmark** (especially when the benchmark is doing well, as in recent history).



Alternatively, we could engineer a portfolio that is designed to excel along both absolute and relative dimensions—but will likely require more flexibility in their governance.

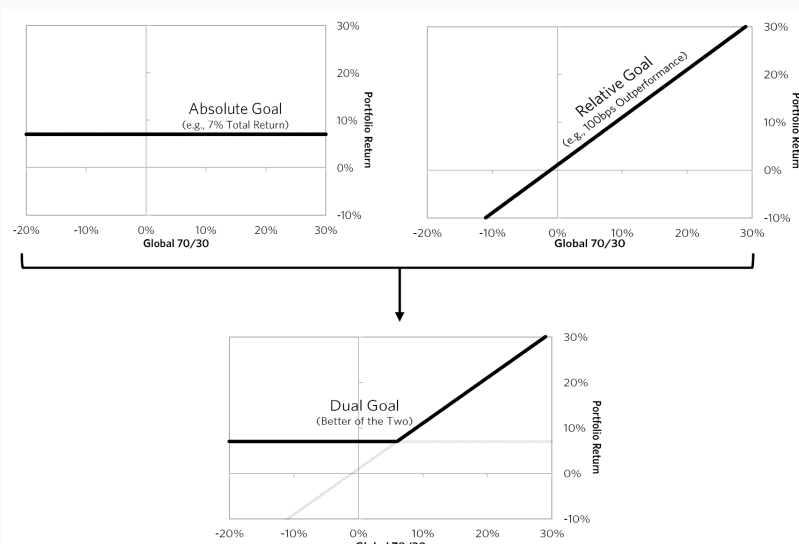


As we’ve wrestled with these trade-offs firsthand, we’ve abstracted a handful of practical points that we find useful and that we think could help investors establish clear goals for portfolio engineering and understand the trade-offs they’ll face given their circumstances.

Taking Stock of the Goal(s)

While some investors are more explicit than others in defining their goals, **the reality is that both absolute and relative returns matter for most**. Most investors have some absolute return need—whether to meet liabilities, build sovereign savings, or preserve enough absolute capital to deploy in the future. At the same time, no investor wants to lag “free” alternatives like a passive market-cap portfolio, and doing so for any extended period of time will likely make for difficult conversations with board members and other stakeholders. In practice, then, most investors have a dual investment goal.

Engineering for dual goals is more complicated than engineering for any single goal, as investment decisions often put the two goals in tension with one another or with other constraints like the amount of leverage or tracking error an investor wants to assume. **A helpful way to represent that goal is not as a single line but as two intersecting lines that provide the option-like profile of dual investment goals.** For example, the charts below show how that illustrative investment objective function described above comes about.



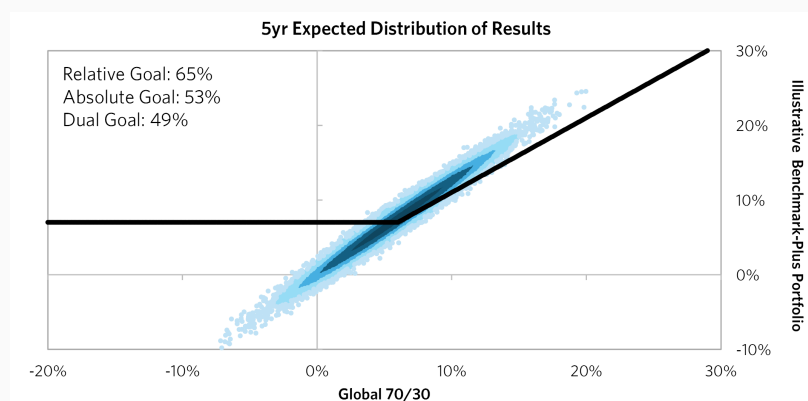
Portfolios can then be assessed against these dual goals, and the principal goal of portfolio engineering becomes shaping a portfolio's return around the objectives you define.

Shaping Returns Toward the Goal(s)

This is where portfolio engineering becomes relevant, because the cumulative effect of all investment decisions determines the distribution of returns relative to an investor's goals. It's possible to optimize the impact of those decisions if they are made explicitly to increase the likelihood of meeting one's goals—a dynamic that can be helpfully visualized in these sorts of return distributions. Those distributions also help clarify some of the trade-offs any investor inevitably faces as they build portfolios around their own unique set of goals and constraints.

The Benchmark-Plus Approach

Take, for example, an investor whose priority is outperforming the market-cap benchmark shown above and who can generate about 1.5% value-add per year with about 3% tracking error. Typically, that investor would aim to build as high an information ratio as possible relative to the benchmark, using whatever investments constitute their opportunity set—public assets, private assets, alternatives, alphas, hedges, etc. **The chart below shows the probabilities that the investor could expect of achieving the dual goals by adding that outperformance and tracking error without fundamentally altering their investment approach**—that is, while remaining highly correlated to the benchmark. These estimates use Bridgewater's long-term capital markets assumptions and show expected results over five years.



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This approach makes significant progress against the relative goal, giving reasonable odds of success over five years. The approach can also be scaled up (through more tracking error, which is relatively easy) or improved (through a higher information ratio, which requires skill) to become even more effective. That said, it's worth remembering how difficult it is to

produce a reliable 0.5 information ratio (i.e., 1.5% outperformance with 3% tracking error).

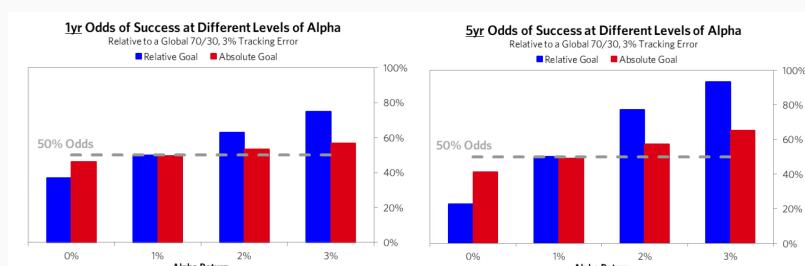
Over much of the last two decades, this approach didn't feel particularly difficult because many ways of seeking additional yield, leverage, or risk played out well (e.g., public-to-private equity shifts, bond-to-credit shifts, doubling down on US concentration, etc.). **These sources of outperformance were highly correlated to most starting portfolios and benchmarks, and they worked out very well. That won't always be the case.**

The difficulty of doing it well notwithstanding, for the investor who only truly cares about relative performance, this approach can work well. In practice, though, that investor is hard to find. Even with realistic-but-quite-good outperformance of the benchmark, we'd estimate that such an investor can expect to miss their 7% absolute return target in about 49% of years, 48% of three-year periods, 47% of five-year periods, and 45% of decades. **Stretching the illustration further, even fantastically good value-add (i.e., a 1.0 information ratio) still leaves an investor exposed to underperformance versus an absolute return goal, particularly over shorter periods.**

How Often Will a Benchmark-Plus Portfolio Miss a 7% Total Return?			
Expectations Using Bridgewater's Long-Term Capital Markets Assumptions			
	With No Skill	With Decent Skill	With Great Skill
	No Alpha, 3% TE	1.5% Alpha, 3% TE	3% Alpha, 3% TE
1 Year	54%	49%	43%
3 Years	57%	48%	38%
5 Years	59%	47%	35%
10 Years	62%	45%	29%

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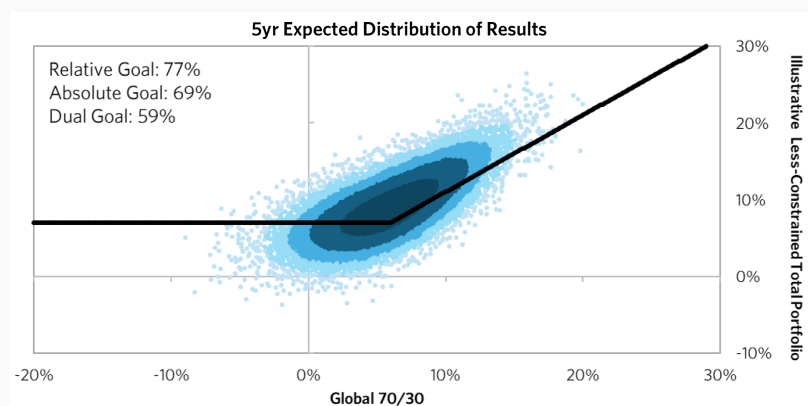
In other words, even where a portfolio proves excellent at beating the benchmark, its outcome will still be more tied to the fate of the market cap than not. Below, we illustrate the probabilities of beating both a relative and an absolute performance goal with different levels of benchmark outperformance. Progress against the absolute return goal is slow and hard-won with this approach, particularly over shorter periods.



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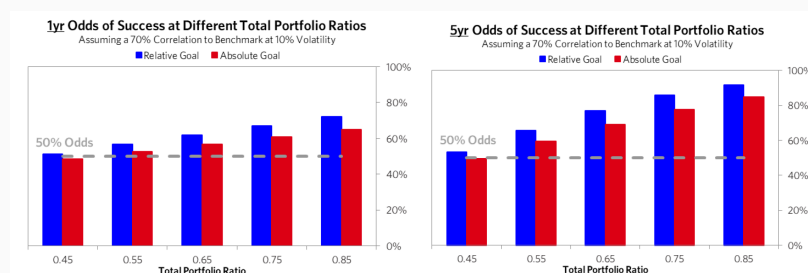
The Total Portfolio Approach

We believe there are ways to vastly improve upon the approach of building from a benchmark. **Consider, instead, starting from a target risk level and then seeking to build the best portfolio at that risk level:** approaching a portfolio in this fundamentally different way gives one the ability to shape a return distribution that looks fundamentally different. Below, we show the returns that can be shaped with this less-constrained approach using another quite-good-but-plausible illustration with a portfolio at 10% volatility and a 0.65 total portfolio return-to-risk ratio.



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Despite not using the benchmark as a reference point at all, this less-constrained approach improves the odds of meeting each goal, with the greatest improvement coming in the odds of achieving the absolute goal. That makes sense given that this approach results in a fundamentally different portfolio, one that need not follow the benchmark down during the most dangerous periods for the market-cap portfolio. Here, too, creating a portfolio with these attributes is not trivial, but it is possible. And while the particulars will vary, **this general approach opens the possibility of improving both absolute and relative performance probabilities simultaneously and more evenly**. Below are the odds that a risk-targeter and ratio-optimizer could give themselves, which you'll notice tend to rise against both goals, particularly as the investment horizon lengthens.



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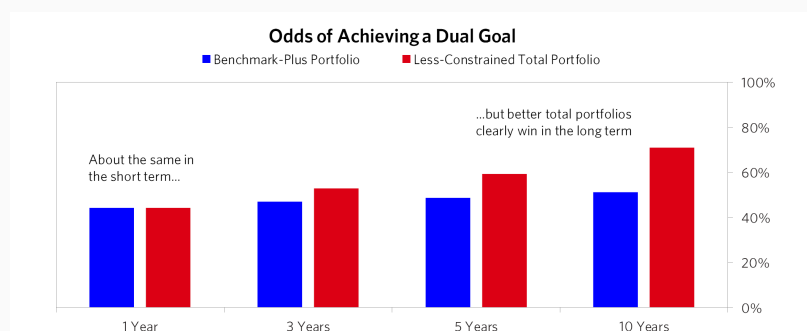
Looking Different by Design

This approach to investing is fundamentally different. That means its outcomes will often look different from the crowd. This is most apparent in measuring the tracking error to a common benchmark, like the global market cap. Below, we show the tracking error generated by building a holistically better portfolio. The example above used a portfolio with a 70% correlation to the global 70/30 and a similar volatility, translating to about a 7% tracking error, a level of tracking error that may be uncomfortably high or constrained.

Expected Tracking Error to a Global 70/30 (Ann)						
		Correlation to 70/30				
		60%	70%	80%	90%	100%
Portfolio Volatility	8%	7.5%	6.5%	5.3%	3.8%	0.7%
	9%	7.9%	6.9%	5.6%	4.0%	0.3%
	10%	8.4%	7.3%	6.0%	4.4%	1.3%
	11%	9.0%	7.9%	6.6%	4.9%	2.3%
	12%	9.7%	8.6%	7.3%	5.6%	3.3%

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Moreover, it means that the success and attractiveness of the Total Portfolio Approach emerges most clearly over the longer term. As shown below, while the odds of achieving dual return goals are roughly the same across the two approaches over short time frames, over 5- or 10-year periods, the better Total Portfolio Approach has significantly higher chances.



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We recognize that less-constrained portfolio construction, which is becoming more popular in TPA-forward organizations, won't fit all investors. But even for those unable to meaningfully change their portfolios, we find these frameworks and analytics powerful for understanding the implicit cost of constraints and having sharper conversations about trade-offs. Having a common language to navigate trade-offs and consider alternatives is helpful, regardless of which decisions are ultimately made.

In a future *Observations*, we'll explore the practical levers available to investors to shape returns toward their goals.

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